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Face recognition

What is face recognition?

Face recognition

[Courtesy of Baidu]

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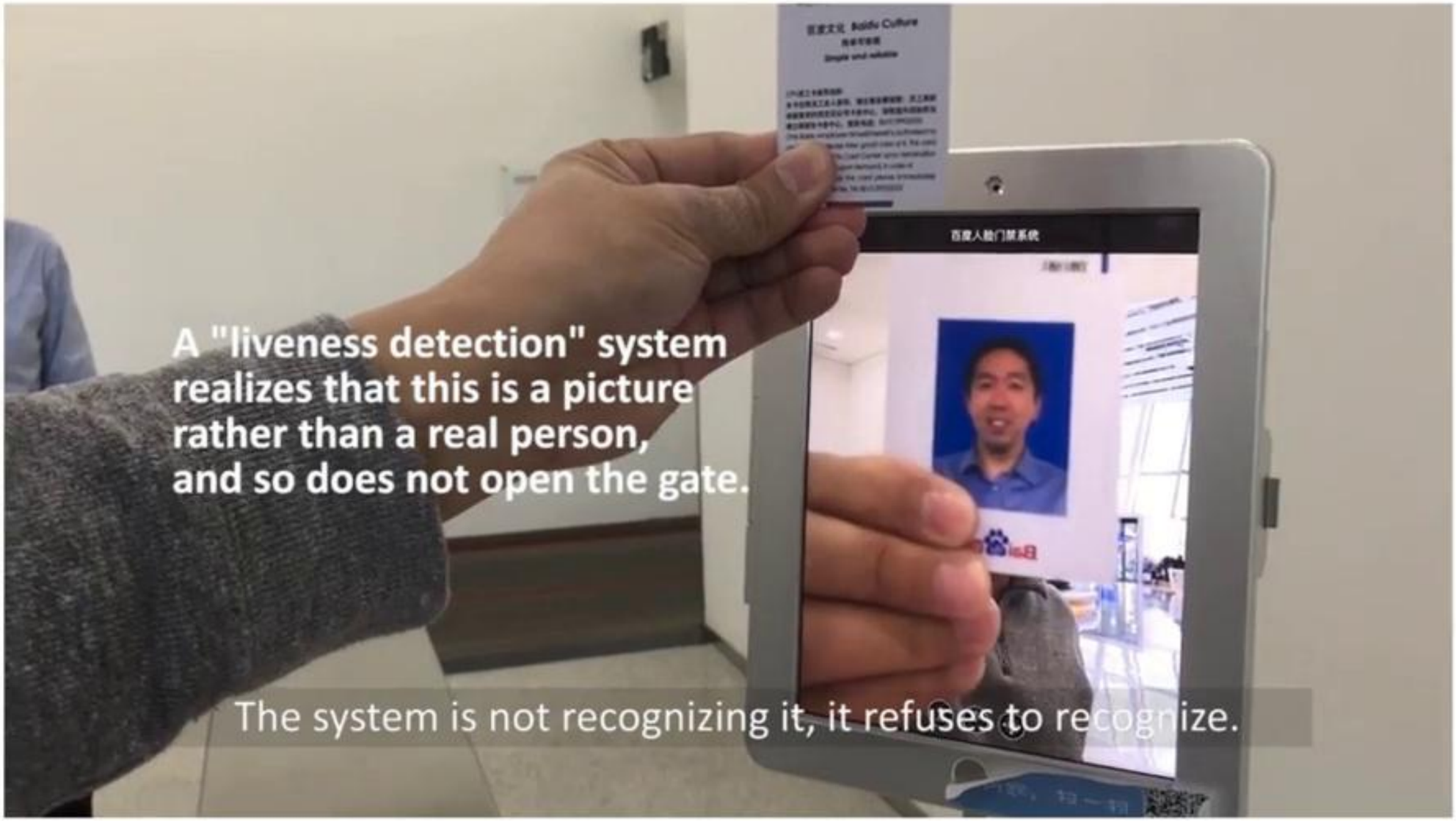
Face recognition



[Courtesy of Baidu]

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Face recognition

[Courtesy of Baidu]

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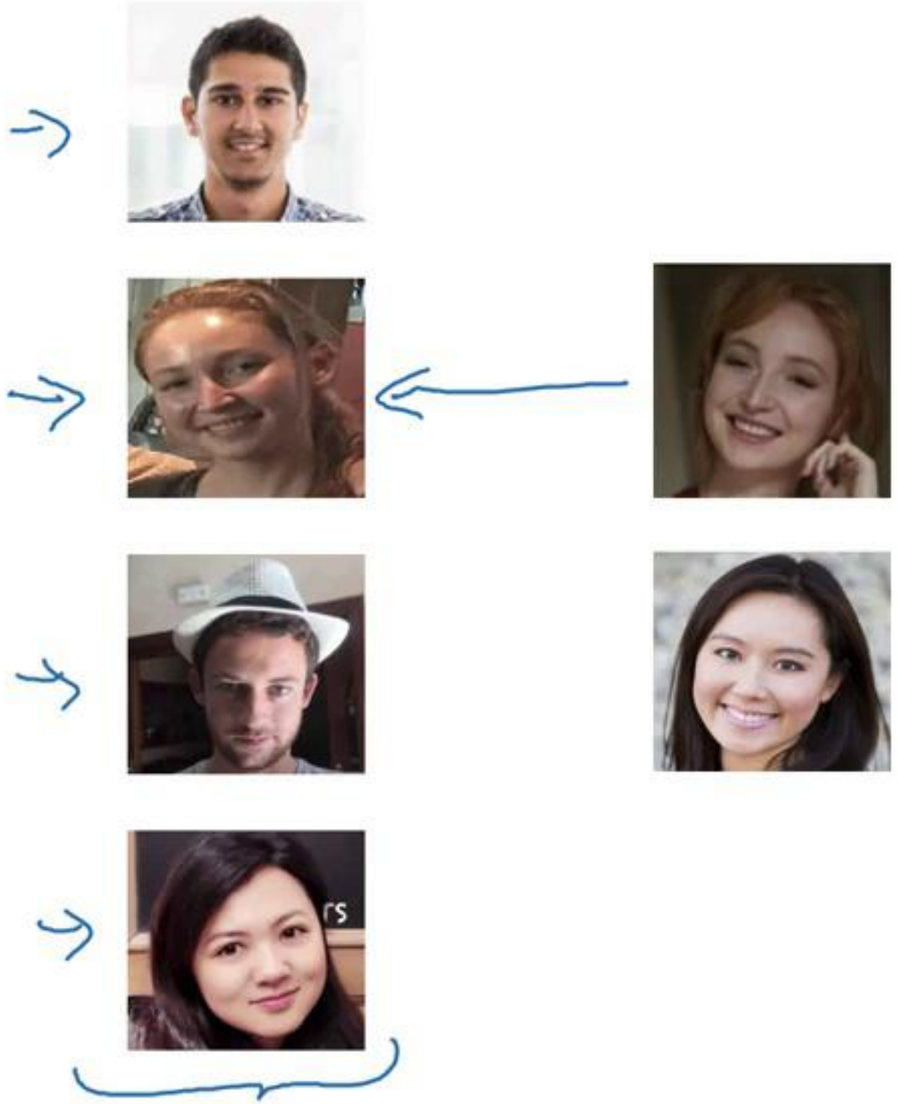
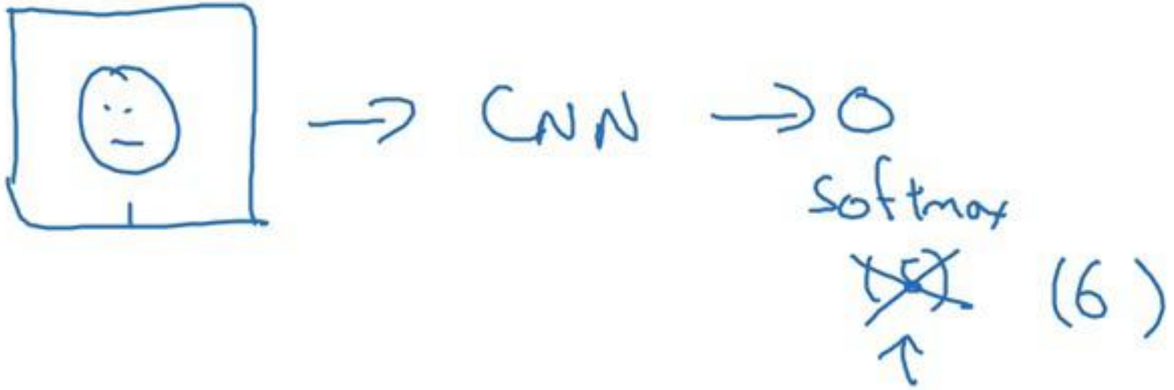
Face recognition

One-shot learning

One-shot learning



Learning from one example to recognize the person again



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Learning a "similarity" function

→ d(img1,img2) = degree of difference between images

If d(img1,img2) $\leq \tau$ "same"

$> \tau$ "different"

} Verification.

d(img1, img2)

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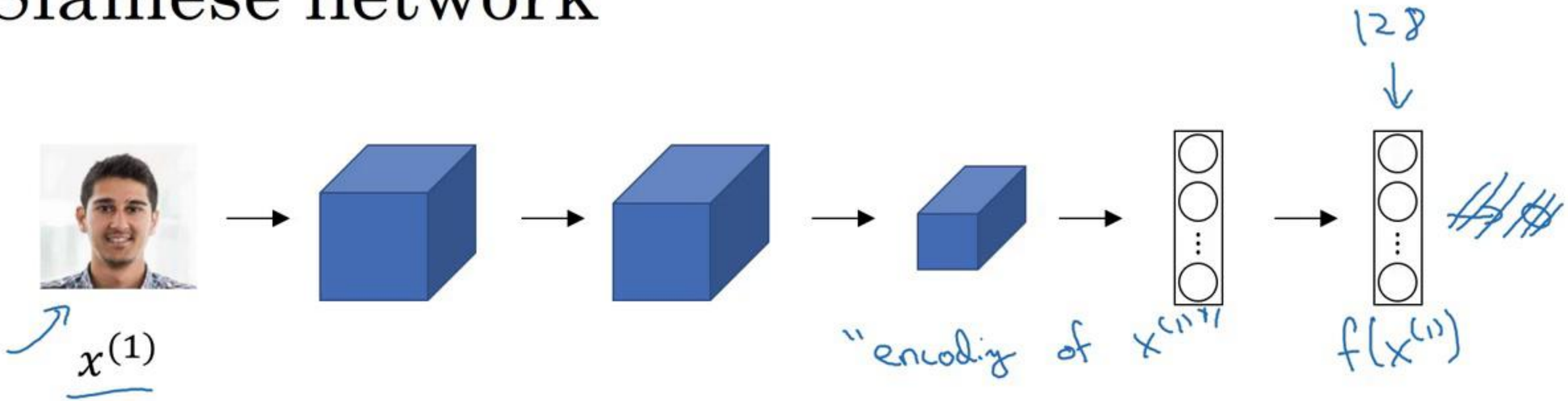
Face recognition

Siamese network

Siamese network



[Taigman et. al., 2014. DeepFace closing the gap to human level performance]

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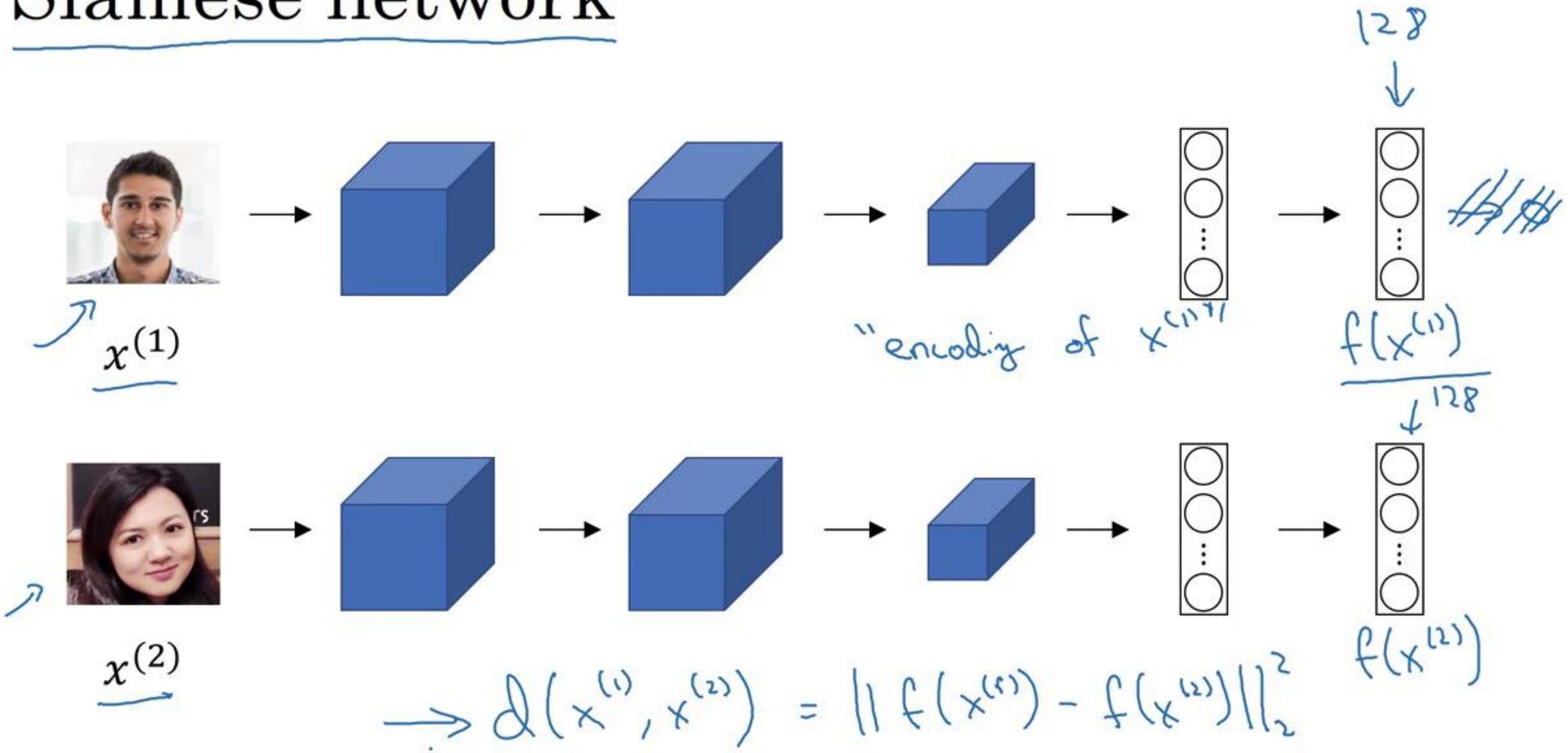
Siamese network



[Taigman et. al., 2014. DeepFace closing the gap to human level performance]

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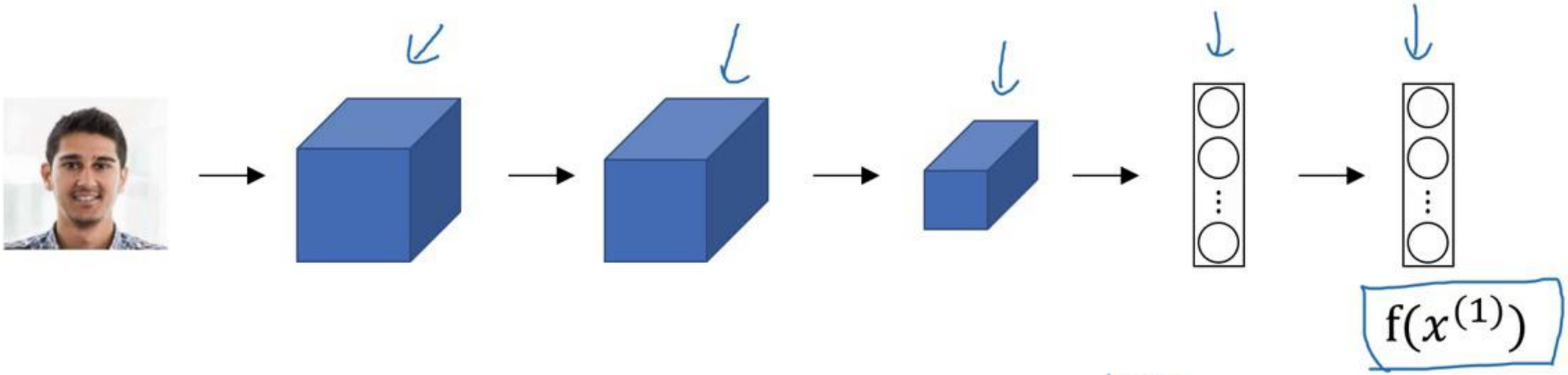
Goal of learning



Parameters of NN define an encoding $f(x^{(i)})$ 128

Learn parameters so that:

If $x^{(i)}, x^{(j)}$ are the same person, $\|f(x^{(i)}) - f(x^{(j)})\|^2$ is small.

If $x^{(i)}, x^{(j)}$ are different persons, $\|f(x^{(i)}) - f(x^{(j)})\|^2$ is large.

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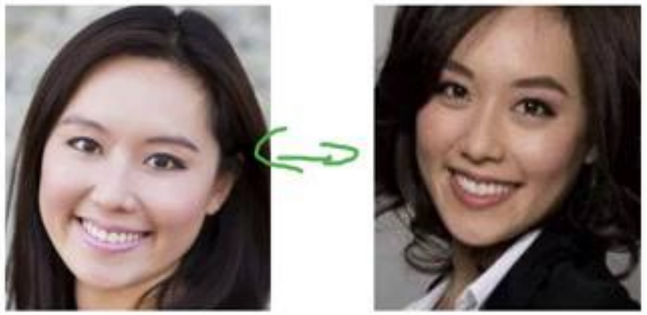
Face recognition

Triplet loss

Learning Objective



Anchor — Positive

Anchor — Negative

A — P: $d(A,P) = 0.5$

A — N: $d(A,N) = \cancel{0.5}\ 0.7$

Want: $\underbrace{\|f(A) - f(P)\|^2}_{d(A,P)} + \alpha \leq \underbrace{\|f(A) - f(N)\|^2}_{d(A,N)}$, $\alpha \to 0.2$

$$\boxed{\underbrace{\|f(A) - f(P)\|^2}_{0} - \underbrace{\|f(A) - f(N)\|^2}_{0} + \underbrace{\alpha}_{\text{margin}} \leq 0}$$

$f(\text{img}) = \vec{0}$

[Schroff et al.,2015, FaceNet: A unified embedding for face recognition and clustering]

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Loss function

Given 3 images A, P, N:

$$\mathcal{L}(A,P,N) = \max\left(\underbrace{\boxed{\|f(A)-f(P)\|^2 - \|f(A)-f(N)\|^2 + \alpha}}_{\leq 0 \quad > 0}, 0\right)$$

$$J = \sum_{i=1}^{m} \mathcal{L}(A^{(i)}, P^{(i)}, N^{(i)})$$

A, P

Training set: 10k pictures of 1k persons

[Schroff et al.,2015, FaceNet: A unified embedding for face recognition and clustering]

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Choosing the triplets A,P,N

During training, if A,P,N are chosen randomly,
$d(A,P) + \alpha \leq d(A,N)$ is easily satisfied.

$\|f(A)-f(P)\|^2 + \alpha \leq \|f(A)-f(N)\|^2$

Choose triplets that're "hard" to train on.

$d(A,P) + \alpha \leq d(A,N)$

$d(A,P) \approx d(A,N)$

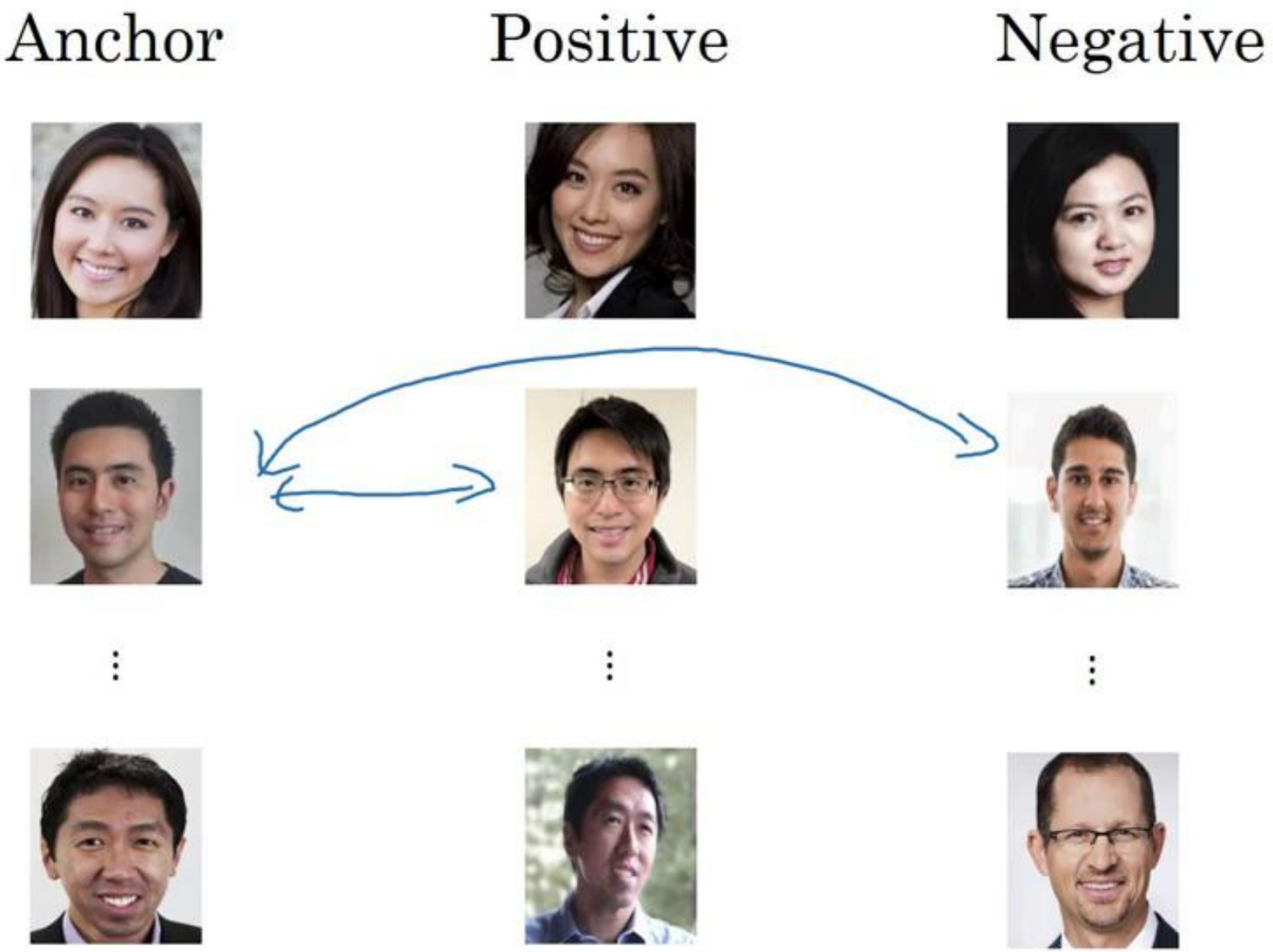
$\downarrow \qquad \uparrow$

Face Net

Deep Face

[Schroff et al.,2015, FaceNet: A unified embedding for face recognition and clustering]

Training set using triplet loss



J

$d(x^{(i)}, x^{(j)})$

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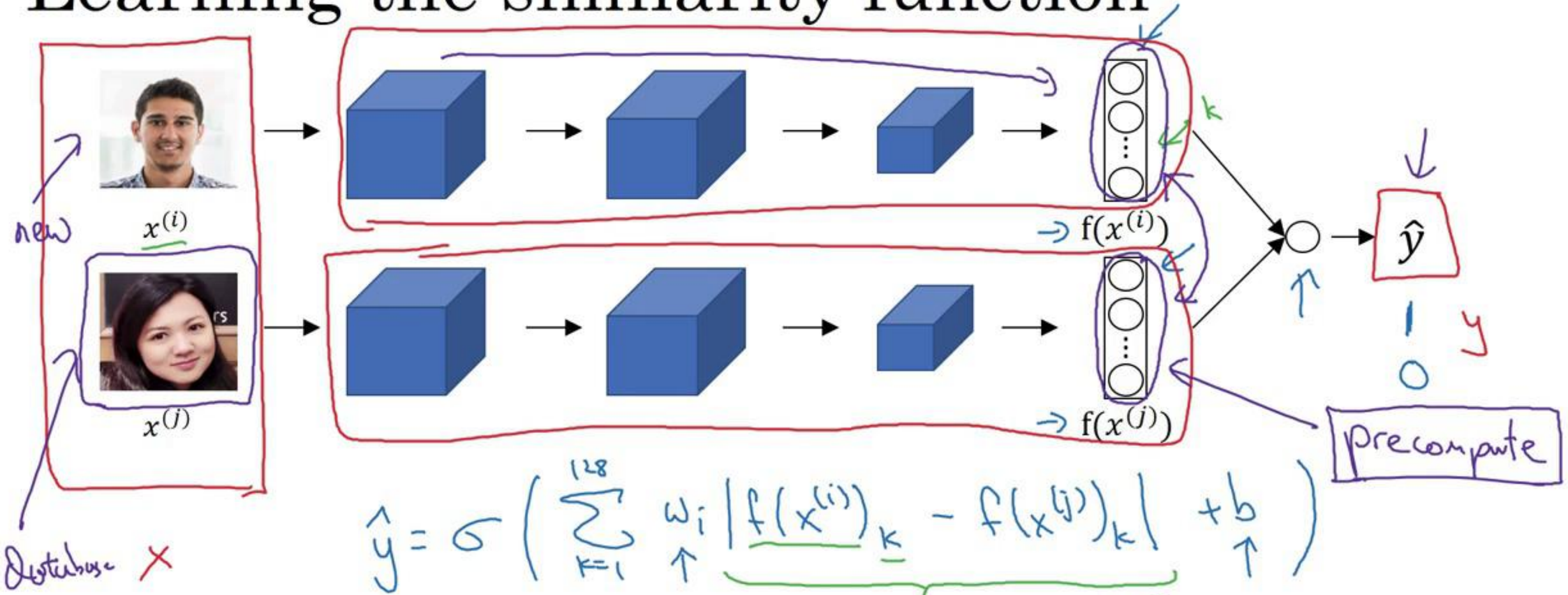
Face recognition

Face verification and binary classification

Learning the similarity function



$$\hat{y} = \sigma\left(\sum_{k=1}^{128} w_i \underbrace{\left|f(x^{(i)})_k - f(x^{(j)})_k\right|}_{\frac{\left(f(x^{(i)})_k - f(x^{(j)})_k\right)^2}{f(x^{(i)})_k + f(x^{(j)})_k}} + b\right)$$

χ^2

[Taigman et. al., 2014. DeepFace closing the gap to human level performance]

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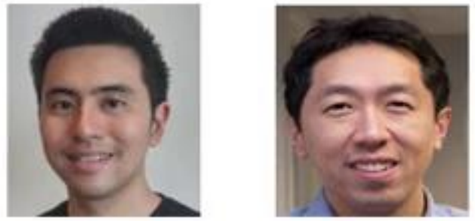
Face verification supervised learning

x	y	
	1	"Same"
	0	"different"
	0	
	1	

[Taigman et. al., 2014. DeepFace closing the gap to human level performance]

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Neural Style Transfer

What is neural style transfer?

Neural style transfer

Content (C)

Style (S)

Generated image (G)

[Images generated by Justin Johnson]

Andrew Ng

Neural style transfer

Content (C) Style (S)

Generated image (G)

Content (C) Style (S)

Generated image (G)

[Images generated by Justin Johnson]

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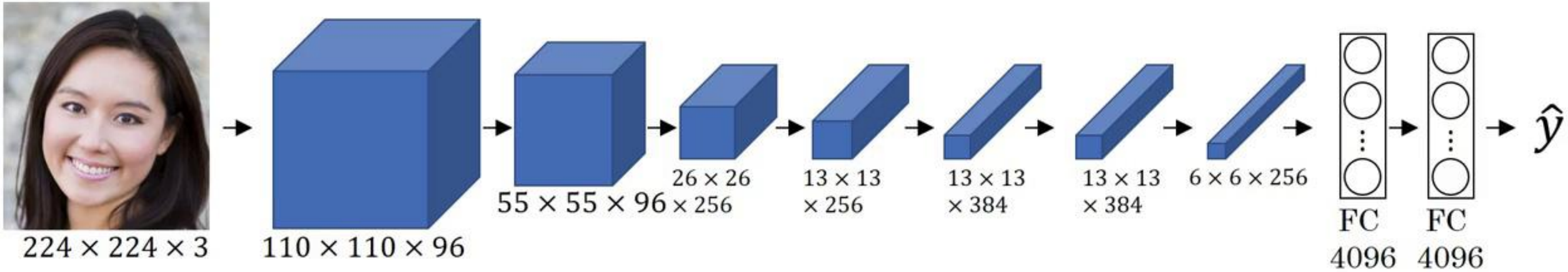
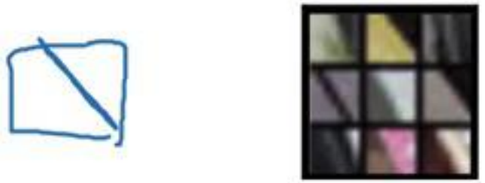
Neural Style Transfer

What are deep ConvNets learning?

Visualizing what a deep network is learning



Pick a unit in layer 1. Find the nine image patches that maximize the unit's activation.



[Zeiler and Fergus., 2013, Visualizing and understanding convolutional networks]

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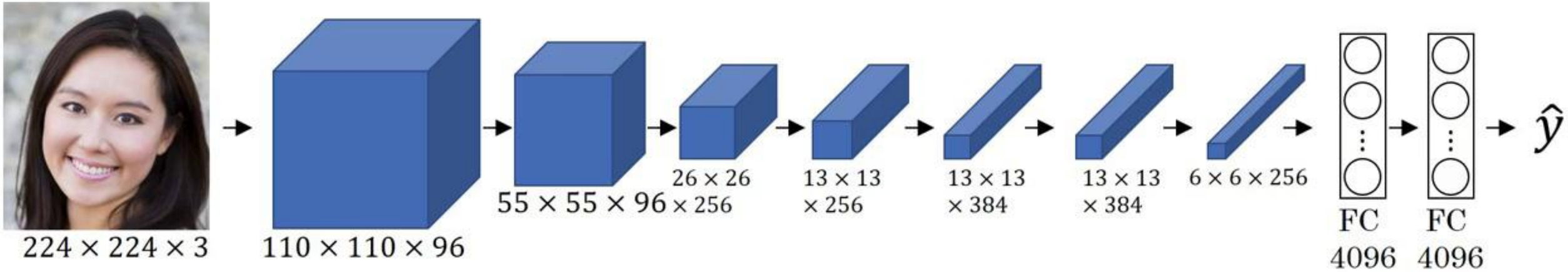
Visualizing what a deep network is learning



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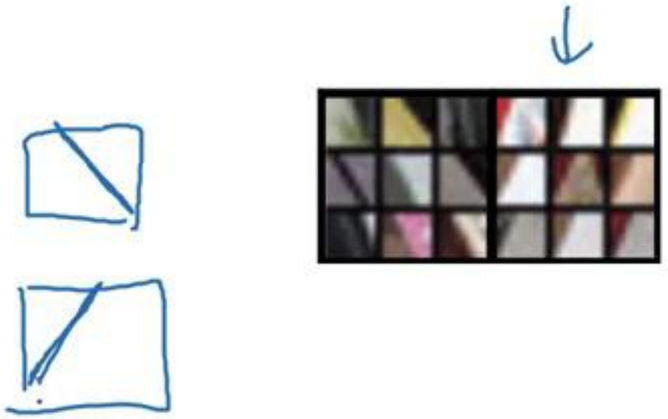
Repeat for other units.



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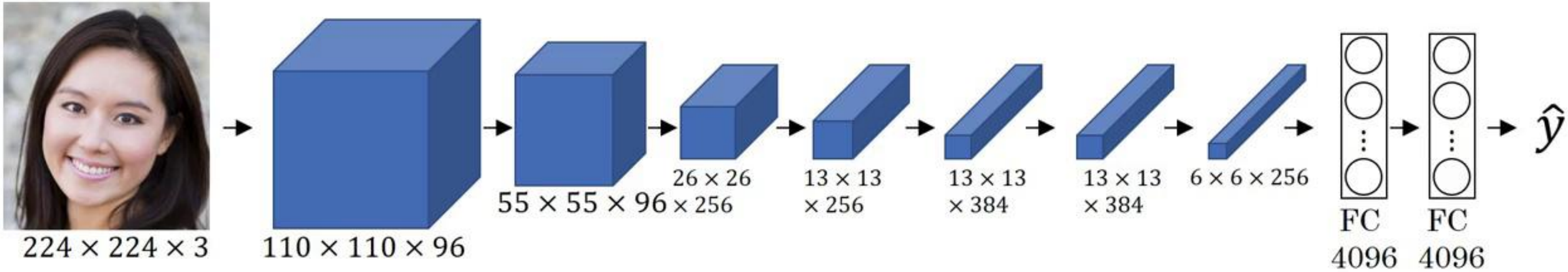

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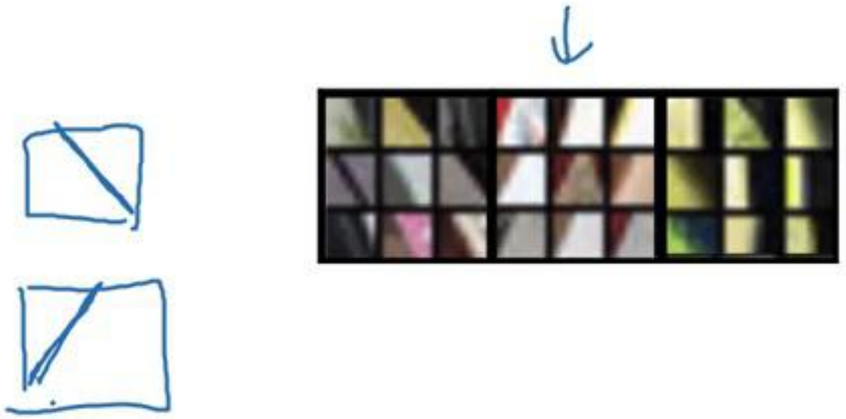

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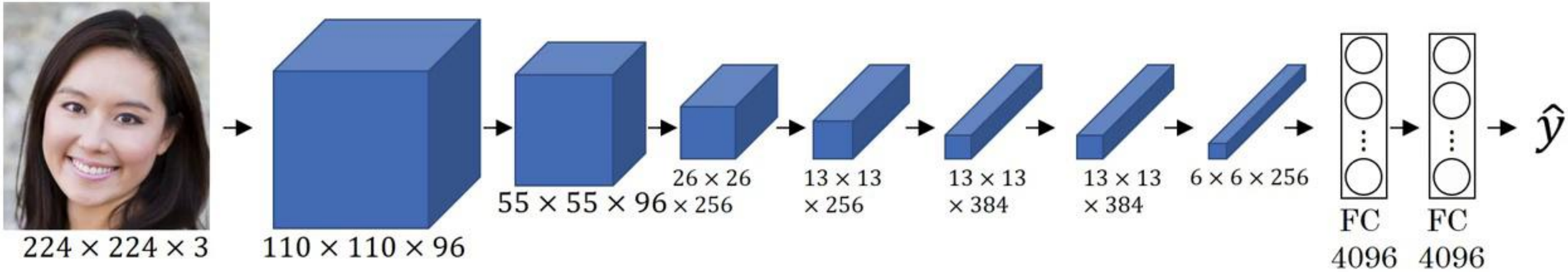

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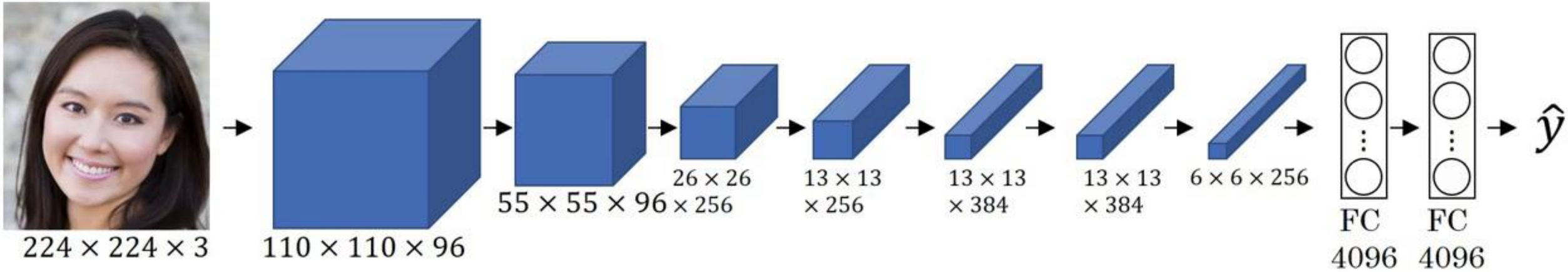

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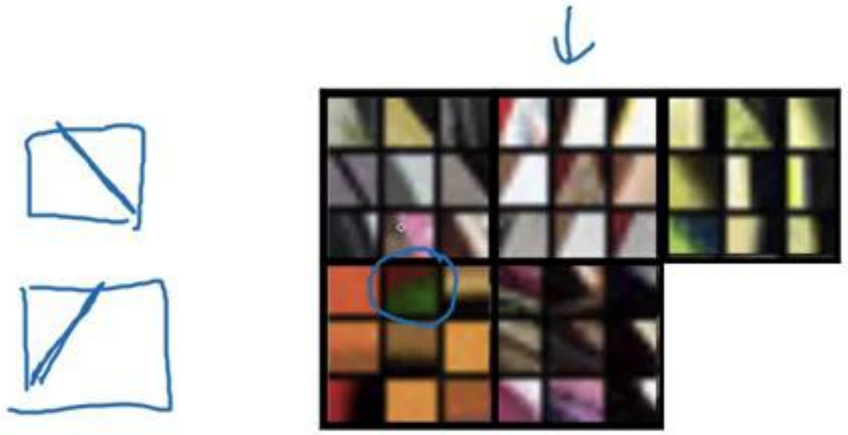

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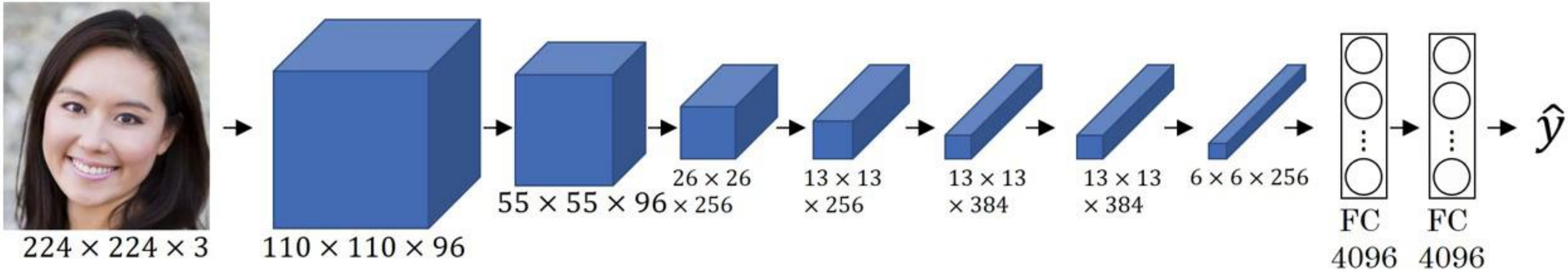

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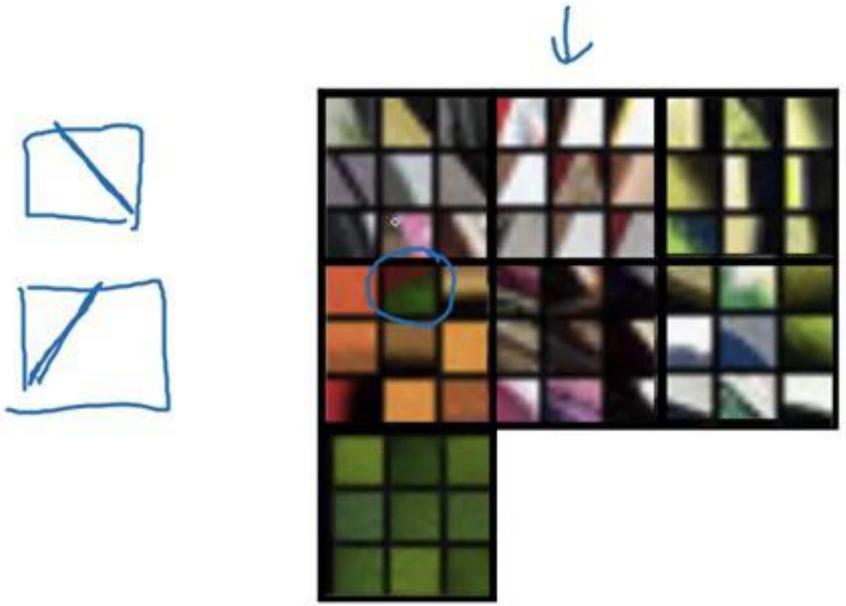

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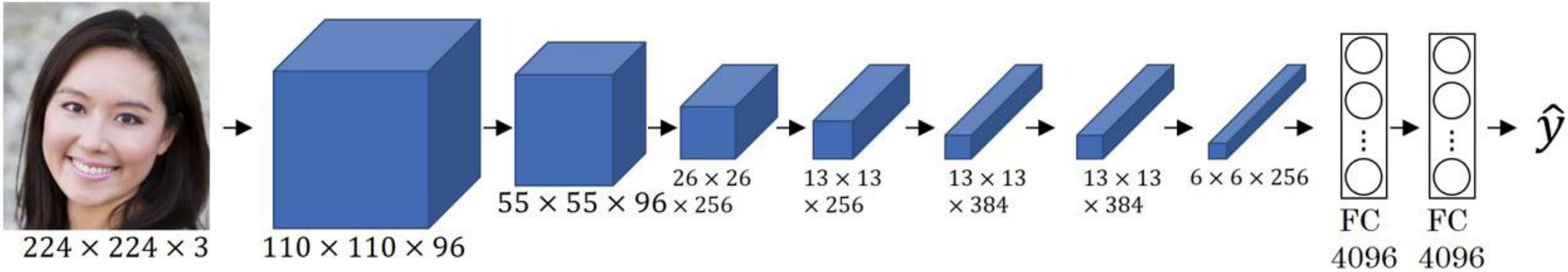

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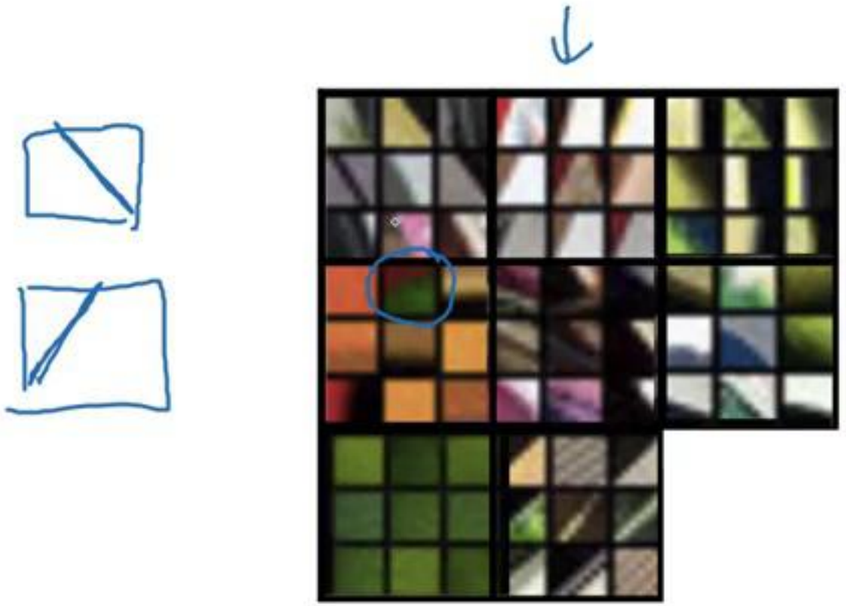

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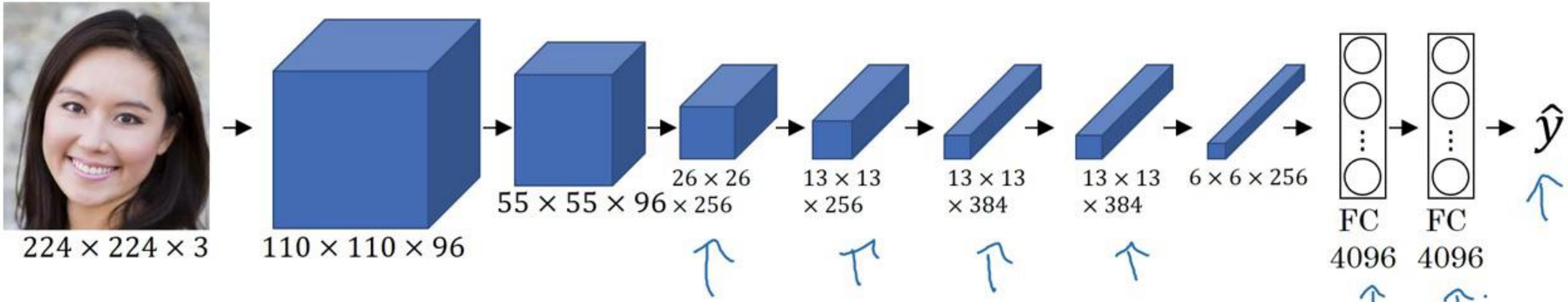

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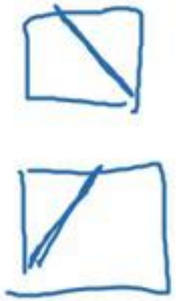

[Zeiler and Fergus., 2013, Visualizing and understanding convolutional networks]

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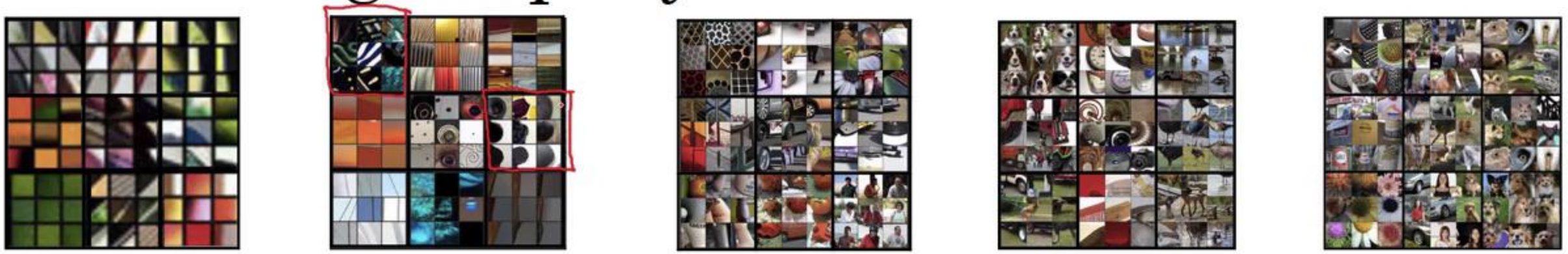
Visualizing deep layers



Layer 1 Layer 2 Layer 3 Layer 4 Layer 5

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Visualizing deep layers: Layer 1

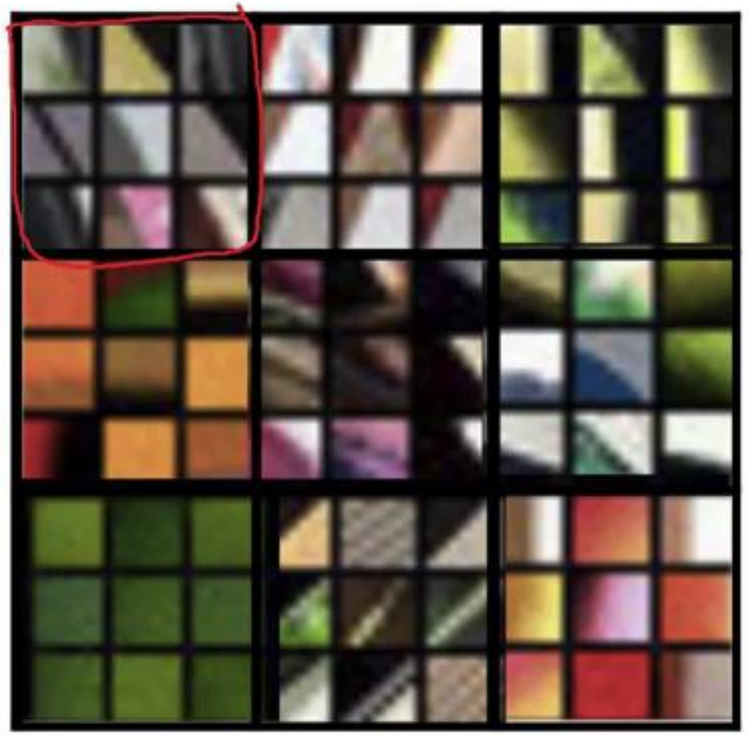
Layer 1

Layer 2

Layer 3

Layer 4

Layer 5



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Visualizing deep layers: Layer 2

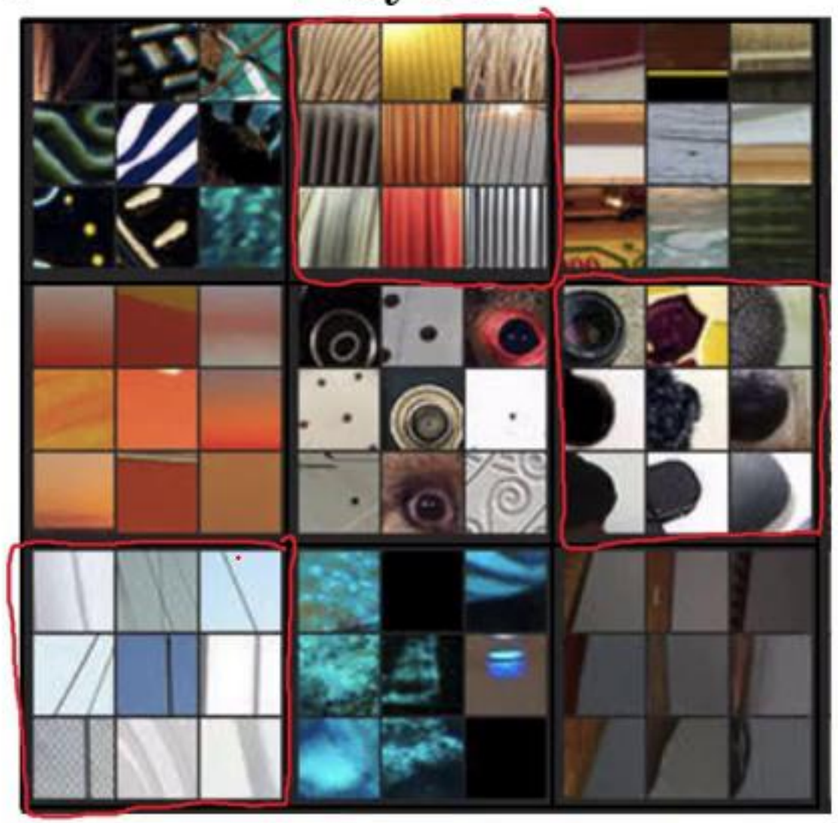
Layer 1

Layer 2

Layer 3

Layer 4

Layer 5



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Visualizing deep layers: Layer 3

Layer 1

Layer 2

Layer 3

Layer 4

Layer 5

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Neural Style Transfer

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Cost function

Neural style transfer cost function

Content C

Style S

Generated image G

$$J(G) = \alpha J_{content}(C, G) + \beta J_{style}(S, G)$$

[Gatys et al., 2015. A neural algorithm of artistic style. Images on slide generated by Justin Johnson]

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Find the generated image G

1. Initiate G randomly

   G: $100 \times 100 \times 3$

   RGB

2. Use gradient descent to minimize $J(G)$

$$G := G - \frac{\partial}{\partial G} J(G)$$

[Gatys et al., 2015. A neural algorithm of artistic style]

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Neural Style Transfer

Content cost function

Content cost function

$$J(G) = \alpha\, J_{content}(C, G) + \beta\, J_{style}(S, G)$$

- Say you use hidden layer l to compute content cost.
- Use pre-trained ConvNet. (E.g., VGG network)
- Let $a^{[l](C)}$ and $a^{[l](G)}$ be the activation of layer l on the images
- If $a^{[l](C)}$ and $a^{[l](G)}$ are similar, both images have similar content

$$J_{content}(C, G) = \frac{1}{2} \underbrace{\| a^{[l](C)} - a^{[l](G)} \|^2}$$

[Gatys et al., 2015. A neural algorithm of artistic style]

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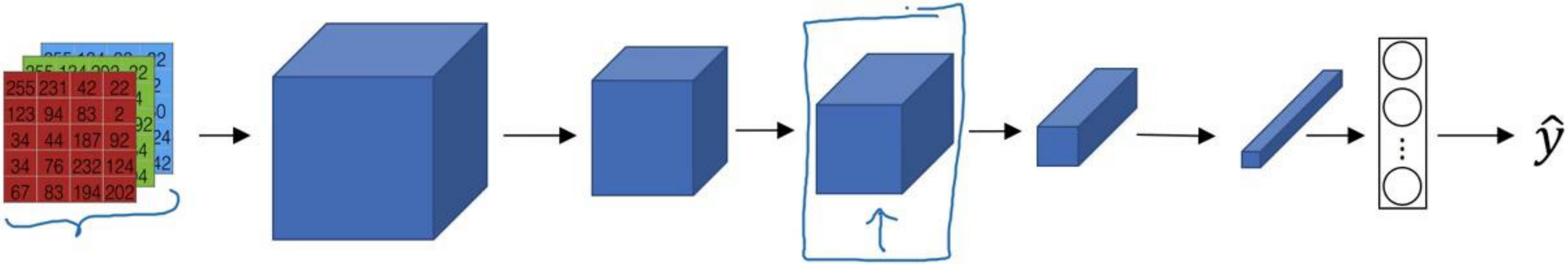
Neural Style Transfer

Style cost function

Meaning of the "style" of an image



Say you are using layer l's activation to measure "style."
Define style as correlation between activations across channels.

[Gatys et al., 2015. A neural algorithm of artistic style]

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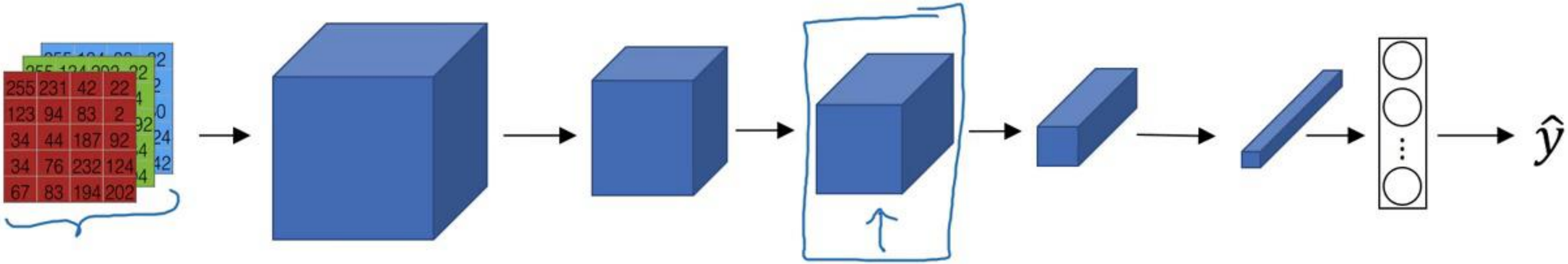
Meaning of the "style" of an image



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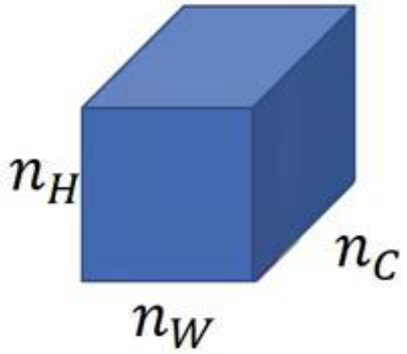

How correlated are the activations across different channels?

[Gatys et al., 2015. A neural algorithm of artistic style]

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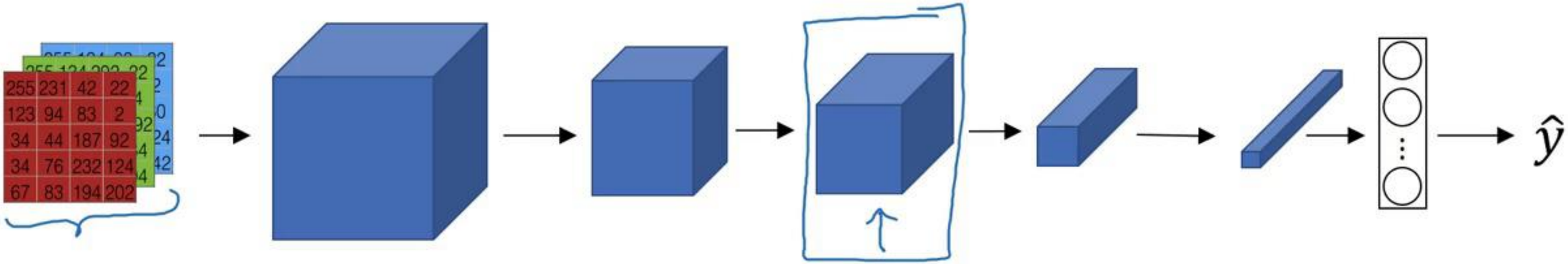
Meaning of the "style" of an image



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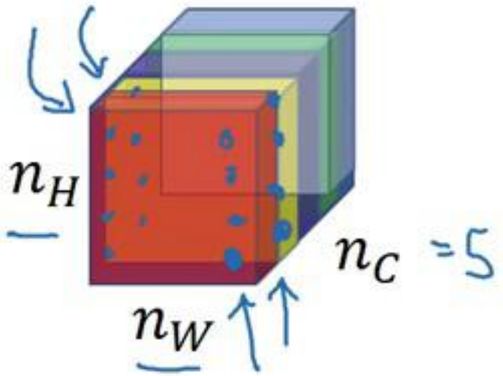

How correlated are the activations across different channels?

[Gatys et al., 2015. A neural algorithm of artistic style]

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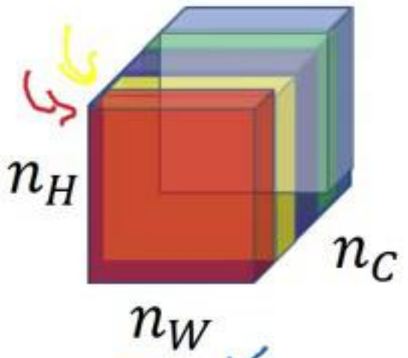
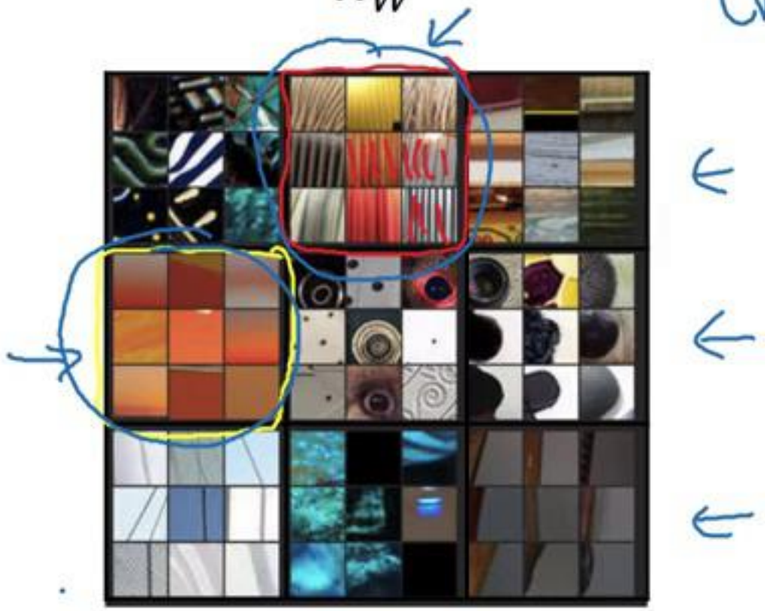
Intuition about style of an image

Style image

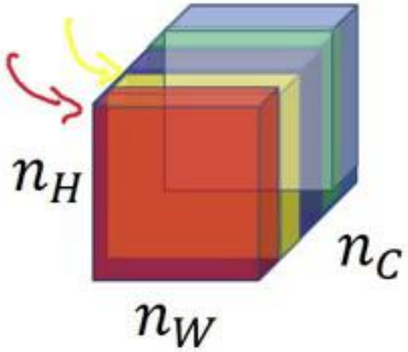


Correlated?

Uncorrelated



Generated Image



[Gatys et al., 2015. A neural algorithm of artistic style]

Andrew Ng

Style matrix

H W C

Let $a^{[l]}_{i,j,k}$ = activation at (i, j, k). $G^{[l]}$ is $n_c^{[l]} \times n_c^{[l]}$

n_c

$G^{[l]}_{kk'}$

$k = 1, \ldots, n_c^{[l]}$

$$\rightarrow G^{[l](S)}_{kk'} = \sum_{i=1}^{n_H^{[l]}} \sum_{j=1}^{n_W^{[l]}} a^{[l](S)}_{ijk} \, a^{[l](S)}_{ijk'}$$

$$\rightarrow G^{[l](G)}_{kk'} = \sum_{i=1}^{n_H^{[l]}} \sum_{j=1}^{n_W^{[l]}} a^{[l](G)}_{ijk} \, a^{[l](G)}_{ijk}$$

"Gram matrix"

$$J^{[l]}_{style}(S, G) = \frac{1}{(\ldots)} \left\| G^{[l](S)} - G^{[l](G)} \right\|_F^2$$

β

$$= \frac{1}{(2 n_H^{[l]} n_W^{[l]} n_c^{[l]})^2} \sum_k \sum_{k'} \left(G^{[l](S)}_{kk'} - G^{[l](G)}_{kk'} \right)^2$$

[Gatys et al., 2015. A neural algorithm of artistic style]

Andrew Ng

Style cost function

$$\| G^{[l](S)} - G^{[l](G)} \|_F^2$$

$$J_{style}^{[l]}(S,G) = \frac{1}{\left(2n_H^{[l]}n_W^{[l]}n_C^{[l]}\right)^2}\sum_k\sum_{k'}\left(G_{kk'}^{[l](S)} - G_{kk'}^{[l](G)}\right)$$

$$J_{style}(S,G) = \sum_l \lambda^{[l]} J_{style}^{[l]}(S,G)$$

$$\underbrace{J(G)}_{G} = \alpha J_{content}(C,G) + \beta J_{style}(S,G)$$

[Gatys et al., 2015. A neural algorithm of artistic style]

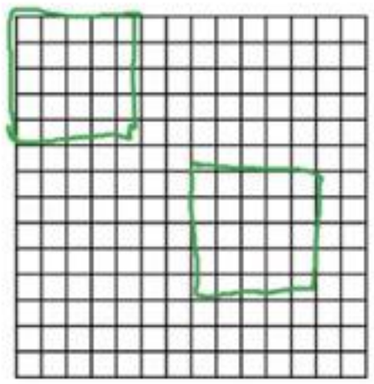
Andrew Ng

Convolutional Networks in 1D or 3D

deeplearning.ai

1D and 3D generalizations of models

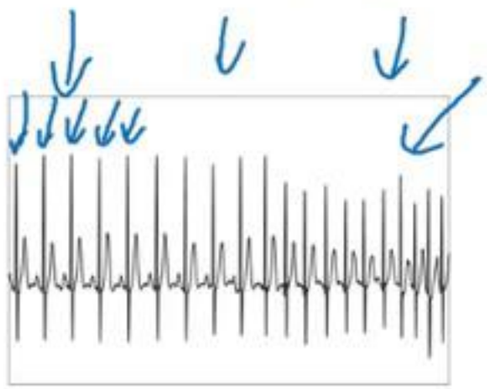
Convolutions in 2D and 1D



2D input image
14×14

$*$

2D filter
5×5



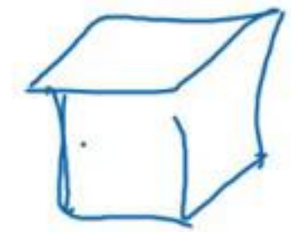
$*$

1	20	15	3	18	12	4	17

14

1	3	10	3	1

5



$14 \times 14 \times \underline{3} \quad * \quad 5 \times 5 \times \underline{3}$

$\rightarrow \underline{10 \times 10} \times 16$

$10 \times 10 \times \underline{16} \quad * \quad 5 \times 5 \times \underline{16}$

$\rightarrow 6 \times 6 \times \underline{32}$

$14 \times \underline{1} \quad * \quad 5 \times \underline{1}$

$\rightarrow 10 \times \underline{16}$

$10 \times \underline{16} \quad * \quad 5 \times \underline{16}$

$\rightarrow 6 * \underline{32}$

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3D data

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3D data

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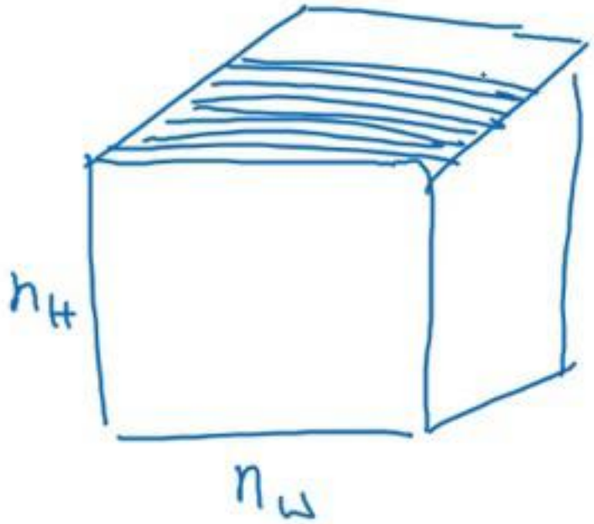
3D data



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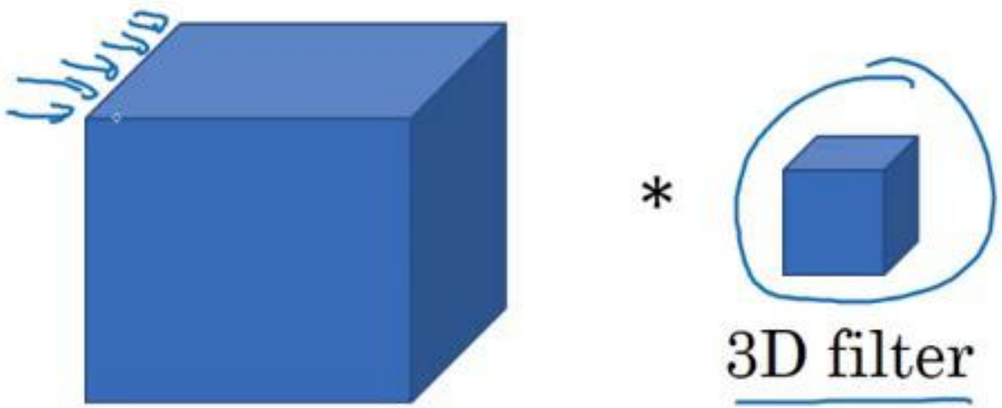
3D convolution



3D volume

n_c

$14 \times 14 \times 14 \times 1$

$* \; 5 \times 5 \times 5 \times 1$

16 filters

$\rightarrow 10 \times 10 \times 10 \times 16$

$* \; 5 \times 5 \times 5 \times 16$

32 filters

$\rightarrow 6 \times 6 \times 6 \times 32$